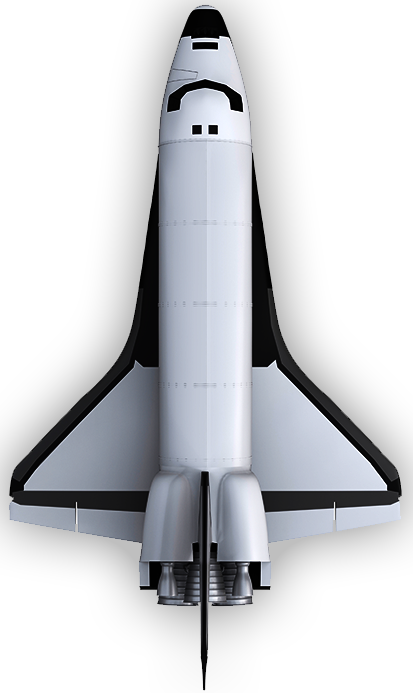
EEEE4064
ADVANCED CONTROL SYSTEM DESIGN

Lecture 6
Observability in State-Space





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Lecture Outline

01 Observability

02 Alternative Forms of Observability

03 Observability in the s-Plane

04 Duality

05 Detectability

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Observability

IMPORTANCE

The concept of observability is useful in solving the problem of reconstructing **unmeasurable** state variables from **measurable** variables in the minimum possible length of time.

APPLICATION

In practice, the difficulty encountered with **state feedback control** is that some of the state variables are not accessible for direct measurement, with the result that it becomes necessary to estimate the unmeasurable state variables in order to construct the control signals.

CONCEPT

The system is **completely observable** if every state $x(t_0)$ can be determined from the observation of $y(t)$ over a finite time interval.

Observability

CONDITION

The system represented by the state space equations is completely observable if and only if the $\mathbf{O}$ matrix is of rank n or has n linearly independent row vectors.

$$\mathbf{O} = \begin{bmatrix} \mathbf{C}^T & \mathbf{A}^T\mathbf{C}^T & \dots & (\mathbf{A}^T)^{n-1}\mathbf{C}^T \end{bmatrix}$$

Observability

EXAMPLE 1

Investigate the state controllability, output controllability, and observability of the following system:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}1 & 1\\ -2 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}0\\ 1\end{bmatrix}u$$

$$y=\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

$\mathbf{M}=\begin{bmatrix}\mathbf{B} & \mathbf{AB}\end{bmatrix}=\begin{bmatrix}0 & 1\\ 1 & -1\end{bmatrix}\Rightarrow$ The rank is 2. Thus, the system is completely state controllable.

$\mathbf{N}=\begin{bmatrix}\mathbf{CB} & \mathbf{CAB}\end{bmatrix}=\begin{bmatrix}0 & 1\end{bmatrix}\Rightarrow$ The rank is 1. Thus, the system is completely output controllable.

$\mathbf{O}=\begin{bmatrix}\mathbf{C}^T & \mathbf{A}^T\mathbf{C}^T\end{bmatrix}=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\Rightarrow$ The rank is 2. Thus, the system is completely observable.

Alternative Forms of Observability

CONCEPT

This approach relies on investigating the observability of systems that are either in the **diagonal** form or the **Jordan** form.

DIAGONAL CONDITION

The condition of complete observability is that if the eigenvectors of $\mathbf{A}$ are distinct, and the matrix $\mathbf{P}^{-1}\mathbf{AP}$ is in the **diagonal** form, then the system is completely observable if none of the columns of the $m \times n$ matrix $\mathbf{CP}$ consists of all zero elements, where:

$$\dot{\mathbf{z}} = \mathbf{P}^{-1}\mathbf{APz} + \mathbf{P}^{-1}\mathbf{B}u, \qquad y = \mathbf{CPz} + \mathbf{D}u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \vdots \\ \dot{z}_n \end{bmatrix} = \begin{bmatrix} -\lambda_1 & 0 & 0 & 0 \\ 0 & -\lambda_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & -\lambda_n \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} + \mathbf{P}^{-1}\mathbf{B}u$$

$$y = \begin{bmatrix} cp_1 & cp_2 & \cdots & cp_n \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} + \mathbf{D}u \qquad \neq 0$$

Alternative Forms of Observability

JORDAN CONDITIONS

The condition of complete observability is that if the eigenvectors of $\mathbf{A}$ are repeated, and the matrix $\mathbf{S}^{-1}\mathbf{AS}$ is in the **Jordan** form, then the system is completely observability if and only if:

1. No two Jordan blocks in $\mathbf{J}$ are associated with the same eigenvalues.
2. No columns of $\mathbf{CS}$ that correspond to the first row of each Jordan block consist of zero elements.
3. The columns of $\mathbf{CS}$ that correspond to distinct eigenvalues are not all zero.

$$\dot{\mathbf{z}} = \mathbf{S}^{-1}\mathbf{ASz} + \mathbf{S}^{-1}\mathbf{B}u, \qquad y = \mathbf{CSz} + \mathbf{D}u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \\ \dot{z}_4 \\ \dot{z}_5 \\ \dot{z}_6 \\ \dot{z}_n \end{bmatrix} = \begin{bmatrix} -\lambda_1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\lambda_1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\lambda_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\lambda_4 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\lambda_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\lambda_5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\lambda_n \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \\ z_5 \\ z_6 \\ z_n \end{bmatrix} + \mathbf{S}^{-1}\mathbf{B}u$$

$$y = \begin{bmatrix} cs_1 & cs_2 & cs_3 & cs_4 & cs_5 & cs_6 & cs_n \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \\ z_5 \\ z_6 \\ z_n \end{bmatrix} + \mathbf{D}u$$

$0 \neq$

Alternative Forms of Observability

EXAMPLE 2

Investigate the observability of the following systems using observation:

1. $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}-1 & 0\\ 0 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix},\quad y=\begin{bmatrix}1 & 3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$

Completely observable.

2. $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}-1 & 0\\ 0 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix},\quad y=\begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$

Not completely observable.

3. $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}2 & 1 & 0\\ 0 & 2 & 1\\ 0 & 0 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix},\quad \begin{bmatrix}y_1\\ y_2\end{bmatrix}=\begin{bmatrix}3 & 0 & 0\\ 4 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$

Completely observable.

Alternative Forms of Observability

4. $$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}2 & 1 & 0\\ 0 & 2 & 1\\ 0 & 0 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix},\quad \begin{bmatrix}y_1\\ y_2\end{bmatrix}=\begin{bmatrix}0 & 1 & 3\\ 0 & 2 & 4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Not completely observable.

5. $$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\\ \dot{x}_4\\ \dot{x}_5\end{bmatrix}=\left[\begin{array}{ccc:cc}2 & 1 & 0 & 0 & 0\\ 0 & 2 & 1 & 0 & 0\\ 0 & 0 & 2 & 0 & 0\\ \hdashline 0 & 0 & 0 & -3 & 1\\ 0 & 0 & 0 & 0 & -3\end{array}\right]\begin{bmatrix}x_1\\ x_2\\ x_3\\ x_4\\ x_5\end{bmatrix},\quad \begin{bmatrix}y_1\\ y_2\end{bmatrix}=\begin{bmatrix}1 & 1 & 1 & 0 & 0\\ 1 & 1 & 1 & 1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\\ x_4\\ x_5\end{bmatrix}$$

Completely observable.

6. $$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\\ \dot{x}_4\\ \dot{x}_5\end{bmatrix}=\left[\begin{array}{ccc:cc}2 & 1 & 0 & 0 & 0\\ 0 & 2 & 1 & 0 & 0\\ 0 & 0 & 2 & 0 & 0\\ \hdashline 0 & 0 & 0 & -3 & 1\\ 0 & 0 & 0 & 0 & -3\end{array}\right]\begin{bmatrix}x_1\\ x_2\\ x_3\\ x_4\\ x_5\end{bmatrix},\quad \begin{bmatrix}y_1\\ y_2\end{bmatrix}=\begin{bmatrix}1 & 1 & 1 & 0 & 0\\ 0 & 1 & 1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\\ x_4\\ x_5\end{bmatrix}$$

Not completely observable.

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Observability in the s-Plane

CONDITION

The necessary and sufficient conditions for complete observability is that **no cancellation** occur in the transfer function or transfer matrix. If cancellation occurs, the canceled mode cannot be observed in the output.

CANCELLATION

- The transfer function has **no cancellation** if and only if the system is completely **state controllable** and completely **observable**.
- This means that the canceled transfer function does not carry along all the **information** characterizing the dynamic system.

Observability in the s-Plane

EXAMPLE 3

Investigate the observability of the following systems using multiple approaches:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -6 & -11 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u$$

$$y=\begin{bmatrix}4 & 5 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

$$\mathbf{O}=\begin{bmatrix}\mathbf{C}^T & \mathbf{A}^T\mathbf{C}^T & (\mathbf{A}^T)^2\mathbf{C}^T\end{bmatrix}=\begin{bmatrix}4 & -6 & 6\\ 5 & -7 & 5\\ 1 & -1 & -1\end{bmatrix}\Rightarrow|\mathbf{O}|=0$$

The system is not completely observable.

Observability in the s-Plane

$$G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D$$

$$G(s) = \begin{bmatrix} 4 & 5 & 1 \end{bmatrix} \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 6 & 11 & s+6 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$G(s) = \frac{s^2 + 5s + 4}{s^3 + 6s^2 + 11s + 6} = \frac{\cancel{(s+1)}(s+4)}{\cancel{(s+1)}(s+2)(s+3)} = \frac{s+4}{(s+2)(s+3)}$$

This confirms that the system is not completely observable.

Duality

CONDITION

The principle of duality states that the system S_1 is completely state controllable if and only if system S_2 is completely observable and vice versa.

$$S_1 = \begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} \end{cases} \qquad S_2 = \begin{cases} \dot{\mathbf{z}} = \mathbf{A}^T\mathbf{z} + \mathbf{C}^T\mathbf{v} \\ \mathbf{n} = \mathbf{B}^T\mathbf{z} \end{cases}$$

IMPORTANCE

The principle of duality expresses the relationship between controllability and observability. The observability of a given system can be checked by testing the state controllability of its dual. Thus, duality allows us to use the same design tools for **observer** problems as for **control** problems with proper substitutions.

Detectability

CONCEPT

- A **fully observable** system is always detectable.
- A **partially observable** system is said to be detectable if all its unobservable modes are stable, regardless of the stability of its observable modes.
- The concept of **detectability** is **dual** to that of **stabilizability**.

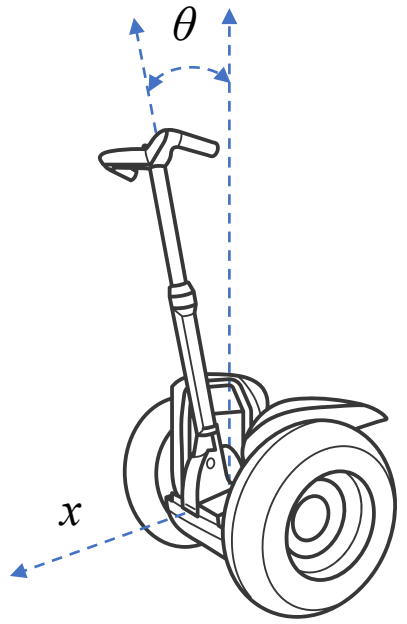
Case Study

EXERCISE

Consider that a company that manufactures segways has approached Prof. Haider Abbas Almurib to help with the design and control of the products. How can you assist Prof. Haider with this task based on what you have studied so far? The state-space representation is as follows:

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -0.107 & 21.344 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -0.047 & 14.106 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ 0.259 \\ 0 \\ 0.114 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}$$



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Controller Design Using Pole Placement

Next Week …

DO YOU HAVE ANY QUESTIONS?